MIL-I-2925B
19 December 1967
SUPERSEDING
MIL-I-2925A
21 November 1960
(See 6.4)

MILITARY SPECIFICATION

INDICATORS, CARBON DIOXIDE, CHEMICAL ABSORPTION TYPE, PORTABLE

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers portable, chemical absorption type, carbon dioxide indicators.

1.2 Classification. The indicators shall be of the following classes, as specified (see 6.2):

Class 1 - For air purification determinations.
Class 2 - For inertness determinations.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

FEDERAL
TT-P-645 - Primer, Paint, Zinc-Chromate, Alkyd Type.
TT-P-664 - Primer Coating, Synthetic, Rust-Inhibiting, Lacquer-Resisting.
PPP-B-636 - Box, Fiberboard.
PPP-B-640 - Boxes, Fiberboard, Corrugated, Triple-Wall.

MILITARY
MIL-P-116 - Preservation, Methods of.
MIL-E-15090 - Enamel, Equipment, Light-Gray (Formula No. 11).

STANDARDS

MILITARY
MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

NATIONAL MOTOR FREIGHT REGULATIONS

(Application for copies should be addressed to the Issuing Officer, National Motor Freight, 1616 P Street, N. W., Washington, D. C. 20036.)

UNIFORM CLASSIFICATION COMMITTEE
Uniform Freight Classification Rules.

(Application for copies should be addressed to the Uniform Classification Committee, 202 Union Station, 516 West Jackson Boulevard, Chicago, Illinois 60606.)

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3. REQUIREMENTS

3.1 Preproduction sample. Prior to beginning production a sample shall be tested as specified in 4.2 (see 6.3).

3.2 Materials. Materials used in the indicators shall be of the best quality for the purpose intended, and shall be of the lightest practicable weight consistent with the strength required for safety, reliability and corrosion resistance.

3.3 Design and construction.

3.3.1 Indicators. Portable absorption carbon dioxide indicators shall be of simple design, rugged construction, and compact. They shall not leak either gas or liquid in any position and shall be spillproof. Indicators shall consist essentially of a saturator, a sampling chamber, a fluid reservoir, and an indication tube. Indicators shall be provided with handles in order that they may be held in such a manner that the heat of the hand will not affect the reading. Indicators shall have sturdy bases.

3.3.2 Saturator. The saturator shall preferably be built into the indicator, which will give the incoming air sample a vapor pressure close to that of the absorption fluid. A tube partially filled with water, and in which the input gas tube opens near the bottom and below the surface of the water will be satisfactory to the command or agency concerned. No packing material shall be used in the saturator.

3.3.3 Sampling chamber. Any required gas sampling device such as an aspirator bulb, will be satisfactory to the command or agency concerned.

3.3.4 Indication tube. The indication tube shall be made of clear transparent plastic. Metal parts shall be of corrosion-resisting material or suitably plated. No discloration of indication tubes shall be evidenced such that it would interfere with the reading of the indicator, at the conclusion of the test specified in 4.6.4.

3.3.5 Absorption fluid. Indicators shall use a single absorption fluid such as a 30 percent potassium hydroxide solution or equivalent, which selectively absorbs carbon-dioxide from the air. The fluid shall be nonflammable, and shall contain no volatile toxic constituents. It may contain, if considered necessary, an antifoaming agent. The refill fluid and the fluid in the indicator shall be the same. Transfer of the indicator from a low to a high ambient temperature shall not affect the accuracy of the indicator as specified in 3.8.1 and 3.9.1, after temperature equilibrium has been attained and the zero of the indicator has been set according to the instructions (see 3.6).

3.4 Calibration of scale. Scales shall be made of corrosion-resisting material. Indicators shall be calibrated by 3 complete calibrations over the entire range of their scale with known mixtures of carbon dioxide and air. At least 3 gas mixtures shall be employed one of which shall have an concentration at or near the maximum determinable by the indicator and the others of which shall be essentially evenly spread between that concentration and zero concentration. At least 3 determinations, the results of which agree within the precision limits specified in 3.8.2 and 3.9.2, shall be made on each gas sample. Each indicator shall be accompanied by a graph showing calibration curves. Indicator scales may be movable so that adjustments of the zero reading may be made, but in no case shall total free movement be more than 0.5 centimeter. Means for avoiding error due to parallax shall be provided.

3.5 Case. Each indicator shall be contained in a metal case and shall not weigh, case included, over 7-1/2 pounds. The case shall be thoroughly cleaned and painted on the inside and outside with one coat of a suitable phosphate treatment, followed by one coat of zinc-chromate primer conforming to the requirements of TT-P-645, or one coat of metal primer conforming to the requirements of TT-P-664. One coat of gray enamel conforming to the requirements of MIL-E-15090 shall be applied over the primer.

3.6 Operating instructions. The operating instructions shall include, explicitly and clearly stated, the method of operating the indicator. They shall contain methods for recalibration, if necessary on replacement of worn parts. They shall include the precautions necessary to secure reproducible results, to detect the exhaustion of the absorption fluid, and to avoid or compensate for the effects of temperature variation, preabsorption of the carbon-dioxide, incorrect zero point, and any other instructions considered necessary by the command or agency concerned for rapid and accurate determinations.

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3.7 Designation and marking.

3.7.1 Identification plates. Identification plates shall be attached to a part of the indicator which will not ordinarily be renewed during its normal service life, and shall be located in a readily accessible position.

3.7.2 Identification plate markings. Data marked on the identification plates shall include the following:

(a) Manufacturer's name, identification, symbols, serial number, contract number, and date of manufacture.
(b) Salient design characteristics; namely, class, range, and similar items.
(c) Blank space for inspector's official stamping.

3.7.3 Case markings. Indicator cases shall be clearly marked "Carbon Dioxide Indicator," together with the range, stock number, manufacturer's name and address, and the contract or order number.

3.8 Class 1 (for air purification determinations).

3.8.1 Range and accuracy. Indicators shall be capable of measuring the carbon dioxide of air over a range of 0 to 5 percent by volume within an accuracy of plus or minus 0.2 percent.

3.8.2 Precision. Indicators when used, as specified in 3.6, shall give reproducible readings within a range of plus or minus 0.15 percent from the average value.

3.8.3 Scale. Indicators shall be graduated in tenths of 1 percent and numbered in unit percent from 0 to 5, inclusive.

3.8.4 Sampling hose. Ten feet of 1/8 inch inside diameter flexible sampling hose shall be furnished as part of the class 1 indicators. The sampling hose shall be made of material which is oil, gas, and vapor resistant and will not absorb or release portions of the gas sample to such an extent as to affect the accuracy of the indicator.

3.9 Class 2 (for inertness determinations).

3.9.1 Range and accuracy. Indicators shall be capable of measuring the carbon dioxide content of tanks containing combustible gases over a range of 0 to 60 percent by volume within an accuracy of plus or minus 2 percent.

3.9.2 Precision. Indicators when used, as specified in 3.6, shall give reproducible readings within a range of plus or minus 1 percent from the average value.

3.9.3 Scale. Indicators shall be graduated in unit percents and shall be marked in multiples of 10 percent from 0 to 60, inclusive.

3.9.4 Sampling hose. Ten feet of 1/8 inch inside diameter flexible sampling hose shall be furnished as part of the class 2 indicators. The sampling hose shall be made of material which is oil, gas, and vapor resistant and will not absorb or release portions of the gas sample to such an extent as to affect the accuracy of the indicator.

3.10 Workmanship. Workmanship shall be first class in every respect. Indicators shall be free from imperfections or defects of manufacture which may affect their appearance, serviceability or operation.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsiblity for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Complete records of the examination and tests shall be kept by the supplier and made available to the Government representative as specified in the contract or order.

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4.2 Preproduction inspection. Preproduction inspection shall consist of the examination and tests specified in 4.4 and 4.6. Production shall not be started until the preproduction inspection report has been approved by the command or agency concerned.

4.3 Quality conformance inspection.

4.3.1 Inspection lot. All indicators offered for delivery at one time shall be considered a lot for the purpose of quality conformance inspection.

4.3.1.1 Certification. The supplier shall furnish written certification with each inspection lot, that he has determined (either by actual test, or through adequate control of the manufacturing process, or by certification by manufacturer), that each type of material meets the specification requirements. Certification shall contain actual test, examination or other verifiable quality data. The proportion of each chemical element shall be shown for comparison with specification. This certification will be acceptable evidence of conformance, unless the quality conformance tests are specified in 4.5 are required and proves nonconformance.

4.3.2 Sampling for examination. Sample indicators shall be selected at random in accordance with MIL-STD-105, at inspection level II, for the examination specified in 4.4.

4.3.3 Sampling for quality conformance tests. Sample indicators shall be selected at random in accordance with MIL-STD-105, at inspection level I, for the tests specified in 4.5.

4.4 Examination. Each of the sample indicators selected in accordance with 4.3.2 shall be examined to verify compliance with this specification. Examination shall be conducted as specified in table I. If any indicator is found not in conformance with the requirements of this specification, this shall be cause for rejection of the lot represented by the sample.

Table I. - Classification of defects

Categories	Defects
Critical:	
1	None defined.
Major:	
101	Indicator not complete, component parts missing.
102	Component parts not new; evidence of reclaimed, used or salvaged components.
103	Evidence of use of unauthorized material or repairs.
104	Indicator not class specified.
105	Indicator not rugged; not compact.
106	Evidence of leakage of gas or liquid; not spill-proof.
107	Handles not furnished; provided, allow heat of hand to affect readings.
108	Indicator does not have sturdy base.
109	Evidence of use of packing material in saturator.
110	Saturator configuration not as specified.
111	Sampling chamber (if required) not as specified.
112	Indication tube not made of clear transparent plastic; metal parts not corrosion-resistant, suitably plated.
113	Absorption fluid not as specified; not non-nonflammable; evidence of volatile toxic constituents.
114	Refill fluid and fluid in indicator not the same. (Not applicable to items packaged with the chambers empty and clean of absorption fluid.)
115	Scales not made of corrosion-resisting material.
116	Indicators not calibrated over entire range; not graduated and numbered as required.
117	Indicator not contained in case; weight including case weighs over maximum allowed.
118	Operating instructions not provided; not clear of explicit methods of operation, methods of recalibration, and operating precautions not included.

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Table I. - Classification of defects (Cont'd)

Categories	Defects
Minor:	
201	Case not thoroughly clean prior to painting; painting not as specified.
202	Identification plates not furnished; not attached to indicator; not permanently marked; not legible; data incomplete.
203	Indicator cases not marked "Carbon Dioxide Indicator"; no range, stock number, manufacturer's name, address, contract or order number missing.

4.5 Quality conformance tests. Each of the sample indicators selected in accordance with 4.3.3 shall be tested in accordance with 4.6 to verify compliance with this specification. If any indicator fails one of more of the tests, this shall be cause for rejection of the lot represented by the sample.

4.6 Test procedures.

4.6.1 Leakage.

4.6.1.1 Gas leakage. A determination shall be made on a gas sample which will give approximately a full scale indication. After the reading has been obtained the indicator shall be allowed to stand without venting for several minutes. A decrease in the reading will result from gas leakage. The temperature of the indicator shall be constant during this test.

4.6.1.2 Fluid leakage. Fluid leakage may be determined by setting the indicator to zero reading by venting and then allowing the indicator to rest in an inverted position for a period of time. This indicator may then be examined for any dripping of fluid or evidence of seepage.

4.6.2 Calibration of scale. Three complete calibrations shall be made of the indicator. At least 3 determinations (the results of which agree within the limits of precision (see 4.6.3)) of the indicator shall be made on each gas sample. On each determination sufficient gas mixtures (at least three) of suitable concentrations to disclose any irregularities in the calibration shall be employed.

4.6.3 Precision. The precision of indicators shall be ascertained by at least 3 concentrations covering the range of the indicator by at least 10 determinations at each of these concentrations.

4.6.4 Serviceability. Serviceability of the indicator and absorbent fluid shall be determined as follows:

(a) Manual operations necessary to perform a determination with the specimen shall be repeated at least 500 times or until the indicator is no longer suitable for use, whichever occurs first. When the utility of the absorbent fluid has been exhausted, it shall be replaced and the operations continued with the refill supply of the absorbent fluid until unsuitability for use is again evidenced. Precautions in the use of the indicator as specified in the instructions (see 3.6) shall be observed before failure shall be attributed to the absorbent fluid or indicator.

(b) Observations shall be made periodically of the condition of all parts of the indicator to determine points of stress, wear, and similar conditions, and minor modifications that may be necessary shall be made and noted. Any discoloration of the indication tube shall be noted.

4.6.5 Chemical and physical analysis. Chemical and physical analysis, as necessary, shall be conducted to determine conformance with the requirements specified herein.

4.7 Possible test failures. Possible test failures shall be determined as follows:

(a) Operational check:
(1) Zero adjustment cannot be quickly and accurately set before each reading.

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(2) Indicators do not measure carbon dioxide content of air over specified range percent by volume.
(3) Indicators do not give readings within specified range percent of average value.

(b) Gas leakage test:
(1) Change in scale indication due to gas leakage.

(c) Fluid leakage test:
(1) Evidence of dripping or seepage of fluid.

(d) Serviceability test:
(1) Evidence of discolor of indication tube.
(2) Excessive wear in parts.
(3) Indicator does not give accurate readings.

4.8 Inspection of preparation for delivery. The preservation, packaging, packing, and marking of the indicator shall be subjected to examination and tests to determine conformance with the requirements of section 5 herein.

5. PREPARATION FOR DELIVERY

(Preparation for delivery requirements specified herein apply only to direct Government procurement. Preparation for delivery requirements between contractors and subcontractors shall be as specified in the individual order.)

5.1 Preservation and packaging.

5.1.1 Level A.

5.1.1.1 Indicators shall be individually preserved and packaged in accordance with method III of MIL-P-116 by placing each indicator within a weather-resistant fiberboard box conforming to type CF, style RSC of PPP-B-636. Cushioning material shall be used to prevent movement of the indicator within the box.

5.1.2 Level C.

5.1.2.1 Preservation and packaging which may be the supplier's commercial practice, shall be sufficient to afford adequate protection against corrosion, deterioration and physical damage during shipment from the supply source to the using activity and until early installation.

5.2 Packing. Packing shall be levels A, B, or C (see 6.2).

5.2.1 Level A.

5.2.1.1 Indicators, preserved and packaged as specified (see 6.2) shall be packed within boxes conforming to PPP-B-640, class 2. The center and all edge seams, including the manufacturer's joint shall be sealed with water-proof pressure-sensitive tape as specified in the appendix of PPP-B-640. In addition, the box shall be banded in accordance with the banding requirements specified in the appendix of the same specification. The gross weight of the box shall not exceed the weight limitations of PPP-B-640.

5.2.2 Level B.

5.2.2.1 Indicators, preserved and packaged as specified (see 6.2), shall be packed in the same manner as specified in 5.2.1, except that the shipping box shall be class 1 of PPP-B-640.

5.2.3 Level C.

5.2.3.1 Indicators preserved and packaged as specified (see 6.2), shall be packed in containers that will insure acceptance by common carrier at the lowest rate and will afford adequate protection against deterioration and physical damage during shipment from the supply source to the first reviewing activity for immediate use. Boxes shall be in compliance with the Uniform Freight Classification rules or National Motor Freight regulations of other common carriers applicable to the mode of transportation.

5.3 Marking. In addition to any special marking required (see 6.2), interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129.

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6. NOTES

6.1 Intended use. The indicators covered by this specification are intended for use of determining the amount of carbon dioxide in air, and the amount of carbon dioxide in tanks containing combustible gases.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Class of indicator (see 1.2).
(c) Preservation packaging, packing or marking required if other than specified in 5.1 (see 5.2 and 5.3).
(d) Stock number.

6.3 Preproduction. Invitations for bids should provide that the Government reserves the right to waive the requirement for preproduction samples as to those bidders offering a product which has been previously procured or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending procurement.

6.4 CHANGES FROM PREVIOUS ISSUE. THE EXTENT OF CHANGES (DELETIONS, ADDITIONS, ETC.) PRECLUDE THE ANNOTATION OF THE INDIVIDUAL CHANGES FROM THE PREVIOUS ISSUE OF THIS DOCUMENT.

Custodians:
Army - MU
Navy - SH

Review activities:
Army - MU
Navy - SH

User activities:
Army - ME
Navy - YD

Preparing activity:
Navy - SH
(Project 6665-0189)

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)*	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

DD FORM 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE
APR 63 (NAVSHIPS 0VPRT 12-66)

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